

Elimination of Cancer Cells in Co-Culture: Role of Different Nanocarriers in Regulation of CD47 and Calreticulin-Induced Phagocytosis.

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Under healthy conditions, pro- and anti-phagocytic signals are balanced. Cluster of Differentiation 47 (CD47) is believed to act as an anti-phagocytic marker that is highly expressed on multiple types of human cancer cells including acute myeloid leukemia (AML) and lung and liver carcinomas, allowing them to escape phagocytosis by macrophages. Downregulating CD47 on cancer cells discloses calreticulin (CRT) to macrophages and recovers their phagocytic activity. Herein, we postulate that using a modified graphene oxide (GO) carrier to deliver small interfering RNA (siRNA) CD47 (CD47_siRNA) in AML, A549 lung, and HepG2 liver cancer cells in co-culture in vitro will silence CD47 and flag cancer cells for CRT-mediated phagocytosis. Results showed a high knockdown efficiency of CD47 and a significant increase in CRT levels simultaneously by using GO formulation as carriers in all used cancer cell lines. The presence of CRT on cancer cells was significantly higher than levels before knockdown of CD47 and was required to achieve phagocytosis in co-culture with human macrophages. Lipid nanoparticles (LNPs) and modified boron nitride nanotubes (BNPs) were used to carry CD47_siRNA, and the knockdown efficiency values of CD47 were compared in three cancer cells in co-culture, with an achieved knockdown efficiency of >95% using LNPs as carriers. Interestingly, the high efficiency of CD47 knockdown was obtained by using the LNPs and BNP carriers; however, an increase in CRT levels on cancer cells was not required for phagocytosis to happen in co-culture with human macrophages, indicating other pathways' involvement in the phagocytosis process. These findings highlight the roles of 2D (graphene oxide), 1D (boron nitride nanotube), and "0D" (lipid nanoparticle) carriers for the delivery of siRNA to eliminate cancer cells in co-culture, likely through different phagocytosis pathways in multiple types of human cancer cells. Moreover, these results provide an explanation of immune therapies that target CD47 and the potential use of these carriers in screening drugs for such therapies in vitro.

MiR-133a/CD47 axis is a novel prognostic biomarker to promote triple negative breast cancer progression.

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Cluster of differentiation 47 (CD47) acts as an anti-engulfment signal on tumor cells, and its overexpression is correlated with poor prognosis of various malignant tumors. However, the role and mechanism of CD47 in tumor cell proliferation, migration and apoptosis remain unclarified. Emerging evidence indicates that microRNAs (miRNAs) are potential regulators to mediate CD47 generation. In this study, we found that CD47 was up-regulated while miR-133a was down-regulated in triple-negative breast cancer (TNBC) in vitro and in vivo. Moreover, we demonstrated for the first time that CD47 was a direct target of miR-133a in TNBC cells, and provided direct evidence of the inverse correlation between miR-133a and CD47 expression in TNBC. Besides, miR-133a functioned as a tumor suppressor to inhibit proliferation and migration, and promote apoptosis of TNBC cells by targeting CD47. Furthermore, overexpression of miR-133a inhibited the tumor growth of TNBC in an in vivo xenograft animal model by targeting CD47. Thus, miR-133a/CD47 axis provides new insight into the mechanism of TNBC progression and could be a promising candidate in the diagnosis and treatment of TNBC.

Lewis Y antigen modified CD47 is an independent risk factor for poor prognosis and promotes early ovarian cancer metastasis.

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CD47 is a membrane receptor that belongs to the immunoglobulin superfamily and plays an important role in the mechanisms of tumor immune escape. CD47 participates in tumor immune escape by combining with SIRP α to reduce the phagocytic activity of macrophages. There are six potential N-glycosylation sites on CD47, and glycosylation is known to be necessary for its membrane localization. However, it is still unknown to what extent glycosylation influences CD47 ligand binding properties and subsequent signaling. By using immunoprecipitation and confocal laser scanning microscopy, we showed that CD47 contains Lewis y antigen. Immunohistochemical analysis demonstrated that both the positive expression and the overexpression of CD47 and Lewis y antigen in cancer tissues and borderline tumors were significantly higher than those in benign ovarian tumors and normal ovarian tissues ($P < 0.05$). A linear correlation between the expression patterns of CD47 and Lewis y antigen was evident ($r = 0.47$, $P < 0.01$). The high expression of CD47 and Lewis y antigen showed significant correlations with the clinical pathological parameters of ovarian cancer [International Federation of Gynecology and Obstetrics (FIGO) standards, lymph node metastasis, and degree of differentiation] ($P < 0.05$). The Cox model and Kaplan-Meier tests showed that high expression of CD47 was an independent adverse risk factor for the prognosis of ovarian cancer. Cases with both high CD47 and Lewis y antigen expression had poor prognoses. Our study demonstrates that Lewis y antigens of CD47 may play a crucial role in the development of ovarian cancer, and could be new targets for immunotherapy for ovarian cancer.

Extracellular Matrix Protein Tenascin C Increases Phagocytosis Mediated by CD47 Loss of Function in Glioblastoma.

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Glioblastomas (GBM) are highly infiltrated by myeloid-derived innate immune cells that contribute to the immunosuppressive nature of the brain tumor microenvironment (TME). CD47 has been shown to mediate immune evasion, as the CD47-SIRP α axis prevents phagocytosis of tumor cells by macrophages and other myeloid cells. In this study, we established CD47 homozygous deletion (CD47 $^{-/-}$) in human and mouse GBM cells and investigated the impact of eliminating the "don't eat me" signal on tumor growth and tumor-TME interactions. CD47 knockout (KO) did not significantly alter tumor cell proliferation in vitro but significantly increased phagocytosis of tumor cells by macrophages in cocultures. Compared with CD47 wild-type xenografts, orthotopic xenografts derived from CD47 $^{-/-}$ tumor cells grew significantly slower with enhanced tumor cell phagocytosis and increased recruitment of M2-like tumor-associated microglia/macrophages (TAM). CD47 KO increased tumor-associated extracellular matrix protein tenascin C (TNC) in xenografts, which was further examined in vitro. CD47 loss of function upregulated TNC expression in tumor cells via a Notch pathway-mediated mechanism. Depletion of TNC in tumor cells enhanced the growth of CD47 $^{-/-}$ xenografts in vivo and decreased the number of TAM. TNC knockdown also inhibited phagocytosis of CD47 $^{-/-}$ tumor cells in cocultures. Furthermore, TNC stimulated release of proinflammatory factors including TNF α via a Toll-like receptor 4 and STAT3-dependent mechanism in human macrophage cells. These results reveal a vital role for TNC in immunomodulation in brain tumor biology and demonstrate the prominence of the TME extracellular matrix in affecting the antitumor function of brain innate immune cells. SIGNIFICANCE: These findings link TNC to CD47-driven phagocytosis and demonstrate that TNC affects the antitumor function of brain TAM, facilitating the

development of novel innate immune system-based therapies for brain tumors.

Clinicopathological and Prognostic Significance of CD47 Expression in Lung Neuroendocrine Tumors.

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Lung neuroendocrine tumors account for approximately 15% of all lung cancer cases. LNET are subdivided into typical carcinoid (TC), atypical carcinoid (AC), large cell neuroendocrine carcinoma (LCNEC), and small-cell lung cancer (SCLC). The Ki-67 index has been used for decades to evaluate mitotic counts however, the role of Ki-67 as a biomarker for assessing prognosis and guiding therapy in metastatic LNET still lacks feasible clinical validation. Recent clinical trials have indicated that inhibition of CD47 with anti-CD47 antibodies exerts a promising antitumor effect against several human malignancies, including NSCLC, melanoma, and hematologic malignancies. However, the clinical relevance of CD47 expression in LNET has remained unclear. We performed a retrospective study in which we analyzed tumor biopsies from 51 patients with a confirmed diagnosis of LNET that received treatment at our hospital. Then, we analyzed if there was any correlation between CD47 expression with any clinical or pathological characteristic. We also analyzed the prognostic significance of CD47, assessed as progression-free survival and overall survival. A total of 51 patients with LNET were enrolled in our study. The mean age at diagnosis was 57.6 (± 11.6) years; 30 patients were women (59%). 27.5% of patients were positive for CD47 expression, and 72.5% of patients showed a CD47 expression of less than 1% and were considered as negatives. In patients with high-grade tumors (this time defined as Ki-67 $> 40\%$), the positive expression of CD47 was strongly associated with an increased PFS. Albeit, these differences did not reach statistical significance when analyzing OS. Contrary to what happens in a wide range of hematologic and solid tumors, a higher expression of CD47 in patients with LNET is associated with a better progression-free survival, especially in patients with a Ki-67 $\geq 40\%$. This "paradox" remains to be confirmed and explained by larger studies.

Galectin-3 Cooperates with CD47 to Suppress Phagocytosis and T-cell Immunity in Gastric Cancer Peritoneal Metastases.

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The peritoneal cavity is a common site of gastric adenocarcinoma (GAC) metastasis. Peritoneal carcinomatosis (PC) is resistant to current therapies and confers poor prognosis, highlighting the need to identify new therapeutic targets. CD47 conveys a "don't eat me" signal to myeloid cells upon binding its receptor signal regulatory protein alpha (SIRP α), which helps tumor cells circumvent macrophage phagocytosis and evade innate immune responses. Previous studies demonstrated that the blockade of CD47 alone results in limited clinical benefits, suggesting that other target(s) might need to be inhibited simultaneously with CD47 to elicit a strong antitumor response. Here, we found that CD47 was highly expressed on malignant PC cells, and elevated CD47 was associated with poor prognosis. Galectin-3 (Gal3) expression correlated with CD47 expression, and coexpression of Gal3 and CD47 was significantly associated with diffuse type, poor differentiation, and tumor relapse. Depletion of Gal3 reduced expression of CD47 through inhibition of c-Myc binding to the CD47 promoter. Furthermore, injection of Gal3-deficient tumor cells into either wild-type and Lgals3 $^{-/-}$ mice led to a reduction in M2 macrophages

and increased T-cell responses compared with Gal3 wild-type tumor cells, indicating that tumor cell-derived Gal3 plays a more important role in GAC progression and phagocytosis than host-derived Gal3. Dual blockade of Gal3 and CD47 collaboratively suppressed tumor growth, increased phagocytosis, repolarized macrophages, and boosted T-cell immune responses. These data uncovered that Gal3 functions together with CD47 to suppress phagocytosis and orchestrate immunosuppression in GAC with PC, which supports exploring a novel combination therapy targeting Gal3 and CD47. Dual inhibition of CD47 and Gal3 enhances tumor cell phagocytosis and reprograms macrophages to overcome the immunosuppressive microenvironment and suppress tumor growth in peritoneal metastasis of gastric adenocarcinoma.

The CD47-SIRP α signaling axis as an innate immune checkpoint in cancer.

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Immune checkpoint inhibitors, including those targeting CTLA-4/B7 and the PD-1/PD-L1 inhibitory pathways, are now available for clinical use in cancer patients, with other interesting checkpoint inhibitors being currently in development. Most of these have the purpose to promote adaptive T cell-mediated immunity against cancer. Here, we review another checkpoint acting to potentiate the activity of innate immune cells towards cancer. This innate immune checkpoint is composed of what has become known as the 'don't-eat me' signal CD47, which is a protein broadly expressed on normal cells and often overexpressed on cancer cells, and its counter-receptor, the myeloid inhibitory immunoreceptor SIRP α . Blocking CD47-SIRP α interactions has been shown to promote the destruction of cancer cells by phagocytes, including macrophages and neutrophils. Furthermore, there is growing evidence that targeting of the CD47-SIRP α axis may also promote antigen-presenting cell function and thereby stimulate adaptive T cell-mediated anti-cancer immunity. The development of CD47-SIRP α checkpoint inhibitors and the potential side effects that these may have are discussed. Collectively, this identifies the CD47-SIRP α axis as a promising innate immune checkpoint in cancer, and with data of the first clinical studies with CD47-SIRP α checkpoint inhibitors expected within the coming years, this is an exciting and rapidly developing field.

Significance of CD47 expression in endometrial carcinoma.

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CD47 is a membrane protein that belongs to the immunoglobulin superfamily and regulates macrophage phagocytosis negatively. As CD47 expression at the cancer cell membrane would inhibit the phagocytic activity of immune cells, it is connected to an unfavorable prognosis in leukemia and malignancies of various solid organs. Materials and. In this study, retrospectively evaluated 72 patients who had been diagnosed with endometrial carcinoma at Pathology Department and had undergone total abdominal hysterectomy and bilateral salpingo-oophorectomy (TAH + BSO) and/or lymphadenectomy. CD47 expression was evaluated in tumorous and nontumor areas in all patients considering cytoplasmic and membranous brown staining in cells. The proportion of expression was evaluated as well as the intensity and an "h score" was obtained. This score was compared with known prognostic parameters. CD47 expressions showed a statistically significant correlation with tumor grade ($P < 0.05$); however, no significant relationship was observed with myometrial invasion depth and lymph vascular invasion status ($P = 0.923$ and $P = 0.754$, respectively). As with other tumors, anti-CD47 antibody may be an alternative

treatment option in patients with high-grade endometrial carcinoma.

Discovery of Anti-CD47 Peptides as Innate Immune Checkpoint Inhibitors.

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Cancer immunotherapy targeting adaptive immune cells has been attracting considerable interest due to its great success in treating multiple cancers. Recently, there is also increasing interest in agents that can stimulate innate immune cell activities. Immune checkpoint inhibitors targeting innate immune cells can block inhibitory interactions ('don't eat me' signals) between tumor cells and phagocytes. CD47 is a transmembrane protein overexpressed in various cancers and acts as a potent 'do not eat me' signal that contributes to the immune evasion of cancer cells. Anti-CD47 peptides that can bind to CD47 and block CD47/SIRP α interaction were discovered using a novel phage display biopanning strategy. Anti-CD47 peptides enhanced the macrophage-mediated phagocytosis of NCI-H82 tumor cells in vitro. Unlike anti-CD47 antibodies, these peptides do not induce the agglutination of RBCs. Moreover, anti-CD47 peptides exhibit high specificity for MC-38 cancer cells expressing CD47. CMP-22 peptide showed the ability to increase the antitumor activity of doxorubicin and extends the survival of CT26 tumor-bearing mice. The discovered anti-CD47 peptides can be considered potential candidates for cancer immunotherapy by blocking the CD47/SIRP α interaction, especially in combination with chemotherapy, to elicit synergistic effects.

CD47 plays a critical role in T-cell recruitment by regulation of LFA-1 and VLA-4 integrin adhesive functions.

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CD47 plays an important but incompletely understood role in the innate and adaptive immune responses. CD47, also called integrin-associated protein, has been demonstrated to associate in cis with $\beta 1$ and $\beta 3$ integrins. Here we test the hypothesis that CD47 regulates adhesive functions of T-cell $\alpha 4\beta 1$ (VLA-4) and $\alpha L\beta 2$ (LFA-1) in in vivo and in vitro models of inflammation. Intravital microscopy studies reveal that CD47(-/-) Th1 cells exhibit reduced interactions with wild-type (WT) inflamed cremaster muscle microvessels. Similarly, murine CD47(-/-) Th1 cells, as compared with WT, showed defects in adhesion and transmigration across tumor necrosis factor- α (TNF- α)-activated murine endothelium and in adhesion to immobilized intercellular adhesion molecule 1 (ICAM-1) and vascular cell adhesion protein 1 (VCAM-1) under flow conditions. Human Jurkat T-cells lacking CD47 also showed reduced adhesion to TNF- α -activated endothelium and ICAM-1 and VCAM-1. In cis interactions between Jurkat T-cell $\beta 2$ integrins and CD47 were detected by fluorescence lifetime imaging microscopy. Unexpectedly, Jurkat CD47 null cells exhibited a striking defect in $\beta 1$ and $\beta 2$ integrin activation in response to Mn(2+) or Mg(2+)/ethylene glycol tetraacetic acid treatment. Our results demonstrate that CD47 associates with $\beta 2$ integrins and is necessary to induce high-affinity conformations of LFA-1 and VLA-4 that recognize their endothelial cell ligands and support leukocyte adhesion and transendothelial migration.